

CS383 - Software Engineering

Software Process Models (II)

Topics to be covered in this lecture

- Software process models in practice
- Agile software development
- IEEE documentations

Software Process Models in Practice

- Software organisations may combine aspects of generic process models to meet the specific needs of their projects
- The choice of process model is also constrained by a range of other factors:
 - Contractual or regulatory requirements
 - Experience and familiarity with the process model
 - Process models used by related projects



Rapid Methodology and overhead problems

- Nowadays software are developed very rapidly
 - Very fast changing of requirements (unstable software!)
 - Software to be evolved quickly to business needs
- Specification, design and implementation are interleaved (combined)
- Systems are developed in series of iterations
- Change user interface using toolset and IDE
- Therefore, using plan-driven approaches may have overheads software process and documentation

Agile Methodology

- Focus on the code rather than design or documentation
- Incremental development of small new releases for clients (usually 2-3 weeks)
- Same as rapid process models, iterative approach
- Software evolves at same speed but maybe with stable versions
- Successful and good for:
 - Small- and medium- sized product
 - Clear commitment of the customer in development process



Principles of Agile Methodology

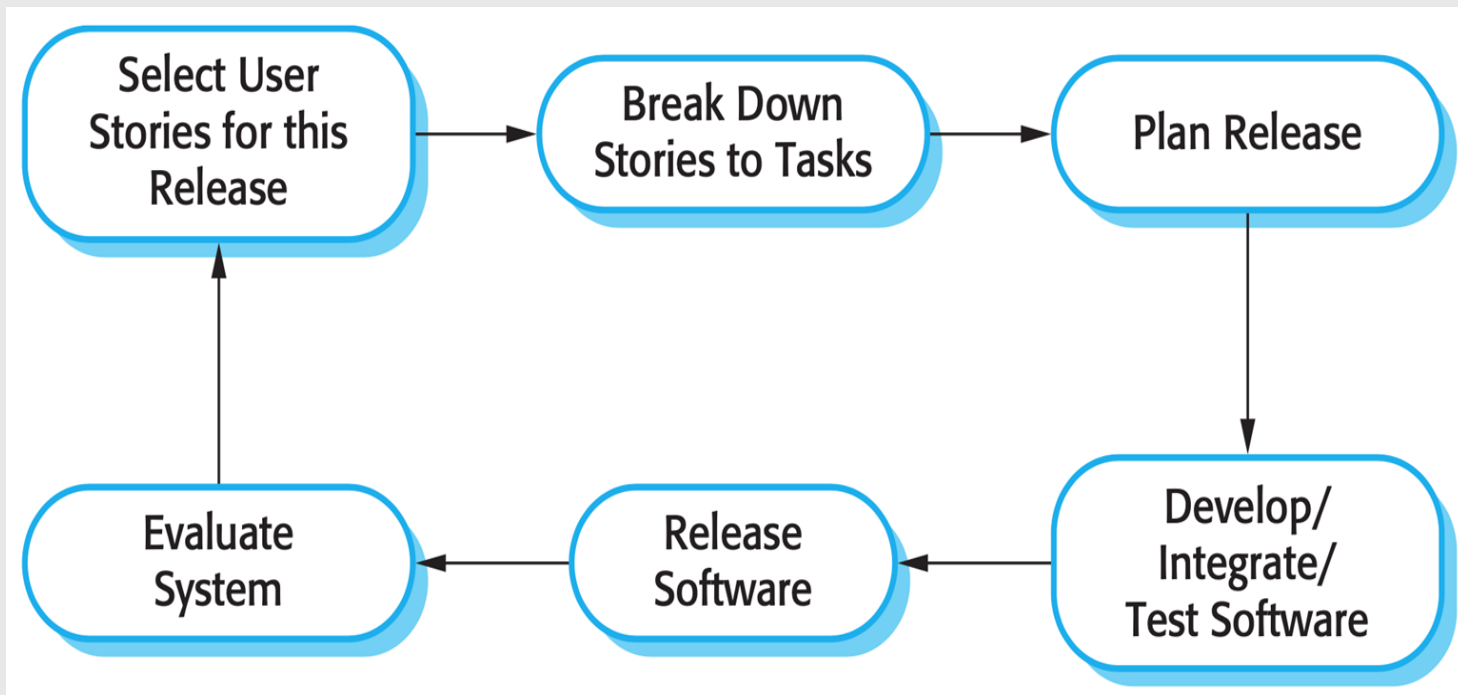
Principle	Description
Customer involvement	Customers should be closely involved throughout the development process. Their role is provide and prioritize new system requirements and to evaluate the iterations of the system.
Incremental delivery	The software is developed in increments with the customer specifying the requirements to be included in each increment.
People not process	The skills of the development team should be recognized and exploited. Team members should be left to develop their own ways of working without prescriptive processes.
Embrace change	Expect the system requirements to change and so design the system to accommodate these changes.
Maintain simplicity	Focus on simplicity in both the software being developed and in the development process. Wherever possible, actively work to eliminate complexity from the system.

Agile Methodology Drawbacks

- Sometimes, practicing agile methods is difficult
 - Difficult to keep the interest of customers
 - Team members may be unsuited to the intense involvement
 - Newly joining team members into the development require documentation to better understand the system
 - Priority changes can be difficult because of multiple stakeholders
 - Maintaining simplicity requires extra work

Agile Method: Extreme Programming

- An 'extreme' approach to iterative development



Agile Method: Extreme Programming (con.t)

1. Incremental planning:

- requirements recorded on story cards and then prioritize
- tasks become the basis of scheduling and cost estimation

2. Small releases:

- frequently and incrementally releases (2-3 weeks window)

3. Simple design:

- just enough design to continue meetings and get requirements

4. Test-first development:

- automated unit test is used for each functionality/task added
(regression testing)

Agile Method: Extreme Programming (cont.d)

5. Refactoring:

- improve code by refactoring more oftenly

6. Pair Programming:

- Developers checking each others code for improvement (understandability)
- Tidy up the code and make it clear (informal review and checks)
- Helps to make refactoring much easier
- Risk to need architecture refactoring for improvement

7. Continuous integration:

- integrate new completed task or functionality into the system (must pass all tests)

8. On-site customer:

- customer should be available full time.

Agile Method: Extreme Programming (cont.d)

- Incremental development (plan-driven) and XP approach (agile)
 - Generally, incremental development tends to degrade the software structure over development time
 - Changes become harder and harder to implement
 - Workaround is applied to overcome hard implementation
 - Usually resulted in duplicate code
 - XP suggests developers do constant refactoring to the code
 - look for possible improvements and implement them immediately

Agile Method: Extreme Programming (cont.d)

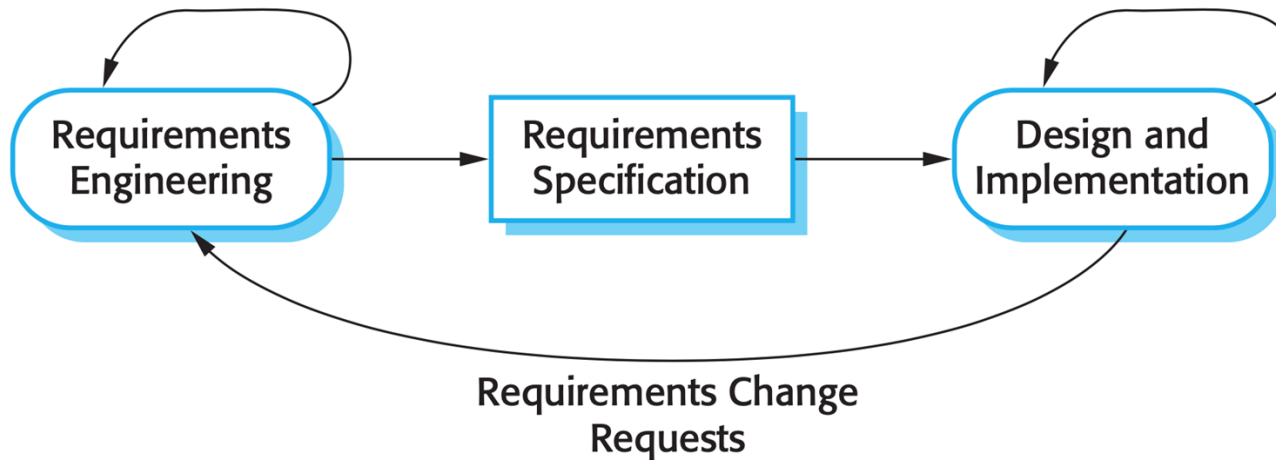
- XP and automated testing
 - Test-first development and automated testing results in a large number of tests
 - However, this does not necessarily lead to *thorough* program testing. Reasons:
 - Programmers sometimes take shortcuts when writing tests
 - Incomplete tests that do not check all variations/complete coverage
 - Not all test can be written incrementally

User Stories

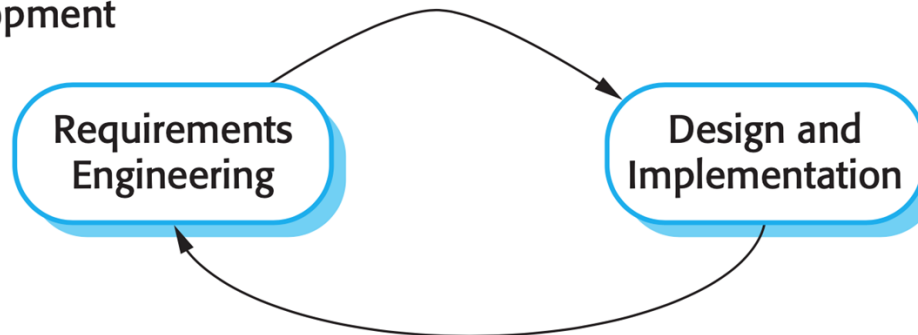
- Project owners and developers do user stories, each story contains
 - Who (user role): customer, employee, admin, etc
 - What (goal): what functionality must be developed
 - Why (reason): why does this functionality needed
- Also, a user story can have estimation of its numerical points
- Examples:
 - As a user, I want a cascade action warning before updating or deleting a record, because I want to avoid dependency problem
 - As a developer, I need a graphical hierarchy model of the developed components updated automatically, so that I better understanding of components dependency.

Plan-Driven and Agile Development

Plan-Based Development



Agile Development



Plan-Driven and Agile Development (cont.)

Concern	Plan Driven	Agile
1- Very detailed specification and design before implementation 2- Deliver software to customers and get rapid realistic feedback 3- Large systems development 4- Systems require a lot of analysis before implementation 5- Long-lifetime system is require 6- Development may require up-to-date technologies for support 7- Development team are distributed (outsourced) 8- Development team has strong rapid development skills 9- The system is subject to regulation/organization (external entity)		

Plan-Driven and Agile Development (cont.)

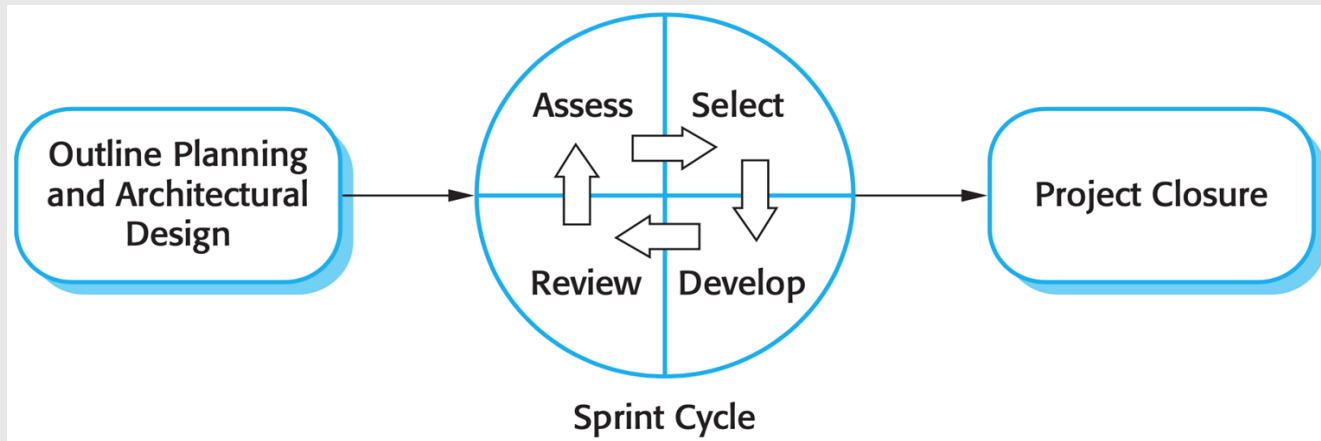
Concern	Plan Driven	Agile
1- Very detailed specification and design before implementation	X	
2- Deliver software to customers and get rapid realistic feedback		X
3- Large systems development	X	
4- Systems require a lot of analysis before implementation	X	
5- Long-lifetime system is require	X	
6- Development may require up-to-date technologies for support		X
7- Development team are distributed (outsourced)	X	
8- Development team has strong rapid development skills		X
9- The system is subject to regulation/organization (external entity)	X	

Agile and project management

- Typical project management work well for plan-driven approaches
 - Clear specification, objectives and documentation are present for management
- In agile, requirements developed incrementally, and new releases are made constantly
- Therefore, different approaches to project management for incremental development and embraces agile are proposed.
- Most notably, Scrum

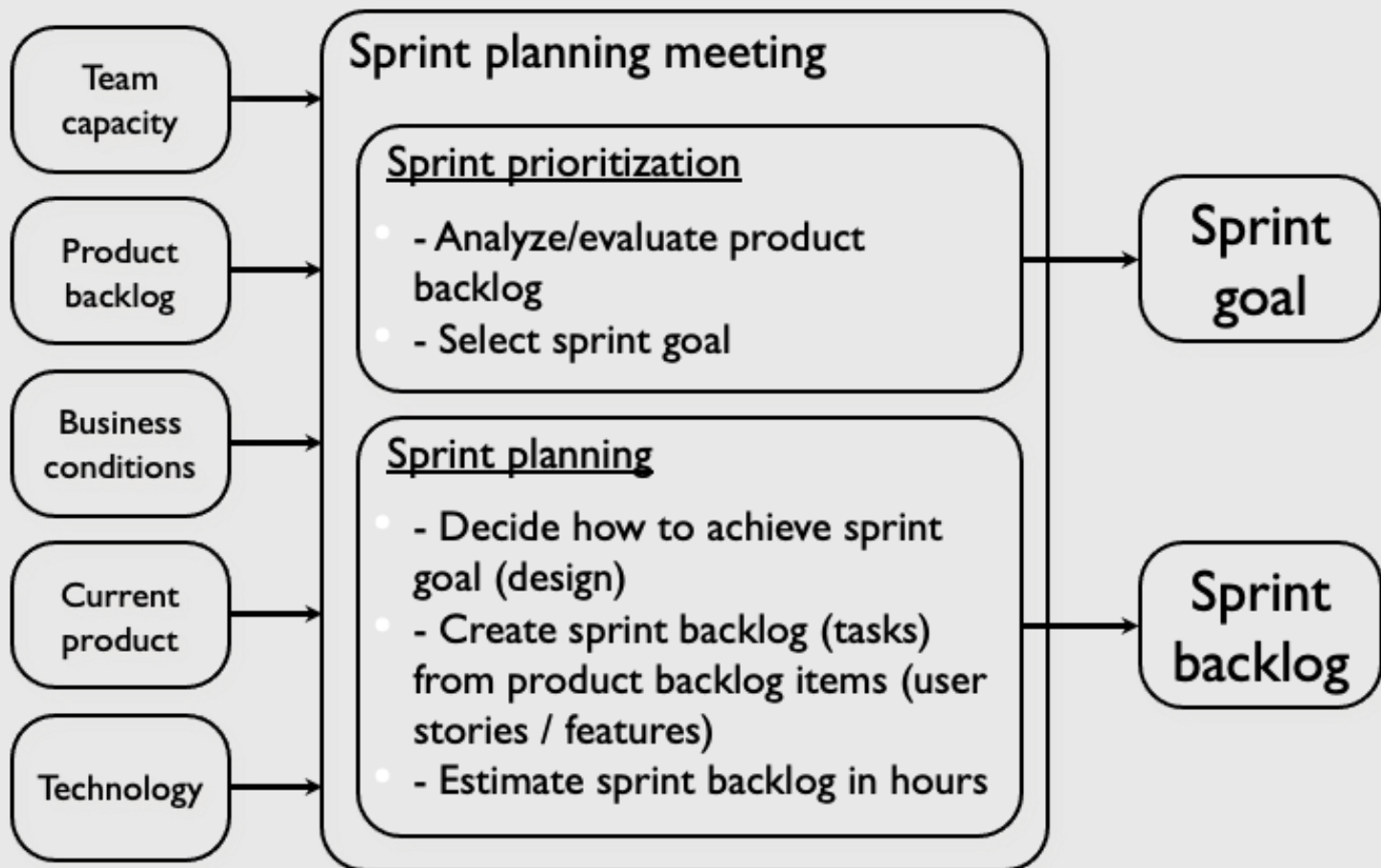
Agile Project Management: Scrum

- Focuses on managing iterative development of agile approach
- Three phases:
 1. The initial phase:
 - Establish general objectives and design the software architecture
 2. Sprint Cycles (2-4 weeks):
 - Each cycle is an increment of the system (features are selected, developed and assessed)
 3. Closure phase:
 - Wrap up the project including documentations



Scrum: sprint planning

- Sprint planning must be established before the sprint cycle



Scrum: sprint planning (cont.d)

- Team capacity: the amount of effort or work that team can put forward to the upcoming sprint.
- Each individual is encourage to be self-organized and determine his/her strength and working/output rate.
 - Then, such detail is used to determine capacity planning of the team

Team capacity = members count x productive days

- Ideally, team capacity is determined after 3rd or 4th sprint cycle (why?)
 - Once we have accurate and actual data of individuals performance
- From team capacity, it is easier to determine Focus Factor
 - Basically, the team's ability to focus on the sprint goals

$$FocusFactor = \frac{SprintVelocity}{TeamCapacity}$$

- Where the sprint velocity is the average completed story points over the past 3 to 5 sprint cycles

Scrum: sprint planning (cont.d)

- **Example:**

- Assume that you have 5 members who are willing to work 8 hrs per day.
- Assume that the average sprint story points (estimated from previous cycles) is 30.

Then the focus factor is:

$$FF = 30 \div (5 \times 8) = 0.75 \text{ (ideal/optimal range is 0.6 to 0.8)}$$

- From focus factor, we can compute the effective team capacity of future sprint cycle.
- Assume that the focus factory is (0.75) and duration of next sprint cycle development is 3 weeks (total 15 days subtracting weekend days), then we have

$$\text{Effective team capacity} = 0.75 \times (5 \times 8 \times 15) = 450 \text{ hours effort estimated}$$

for next sprint

Example of Product Backlog

ID	Story	Estimation	Priority
7	As an unauthorized User I want to create a new account	3	1
1	As an unauthorized User I want to login	1	2
10	As an authorized User I want to logout	1	3
9	Create script to purge database	1	4
2	As an authorized User I want to see the list of items so that I can select one	2	5
4	As an authorized User I want to add a new item so that it appears in the list	5	6
3	As an authorized User I want to delete the selected item	2	7
5	As an authorized User I want to edit the selected item	5	8
6	As an authorized User I want to set a reminder for a selected item so that I am reminded when item is due	8	9
8	As an administrator I want to see the list of accounts on login	2	10
Total		30	

Example of Sprint Backlog

Tasks	Mon	Tue	Wed	Thu	Fri
Code the user interface	8	4	8		
Code the middle tier	16	12	10	4	
Test the middle tier	8	16	16	11	8
Write online help	12				
Write the Foo class	8	8	8	8	8
Add error logging			8	4	

Summary: Software Process Model

	Waterfall model	Incremental development	Reuse-oriented development	Agile methods
Requirements Volatility	Low	Medium	Low	Hight
Project Size	Large	Medium	Any	Small
Customer Involvement	Low	Medium	Low	Hight
Technical Risk	Low	High	Medium	High
Release Schedule	Long	Medium	Short	Very Short